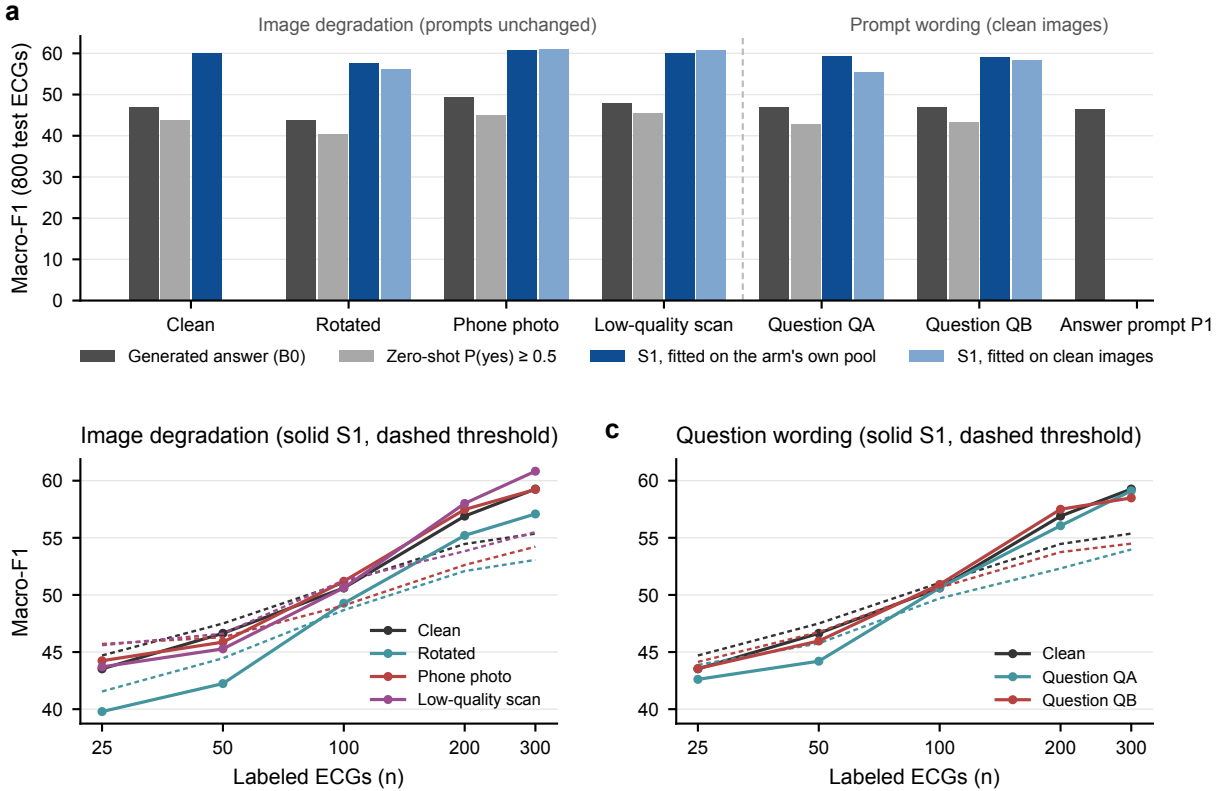
a
Image degradation (prompts unchanged)
Prompt wording (clean images)
Macro-F1 (800 test ECGs)
0
10
20
30
40
50
60
Clean
Rotated
Phone photo
Low-quality scan
Question QA
Question QB
Answer prompt P1
Generated answer (B0)
Zero-shot P(yes) ≥ 0.5
S1, fitted on the arm's own pool
S1, fitted on clean images
b
Image degradation (solid S1, dashed threshold)
Macro-F1
40
45
50
55
60
25
50
100
200
300
Labeled ECGs (n)
Clean
Rotated
Phone photo
Low-quality scan
c
Question wording (solid S1, dashed threshold)
Macro-F1
40
45
50
55
60
25
50
100
200
300
Labeled ECGs (n)
Clean
Question QA
Question QB